

More AI Reviewers Don't Help — Decorrelation Does

A pre-registered, compute-matched comparison of four AI code-review architectures across **2,415 reviews** — including 30 real pull requests from the platform we built for our client.

Companion to: “Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study”

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THE CONTRIBUTION

Agreement predicts truth — but **only** when the reviewers are independent.

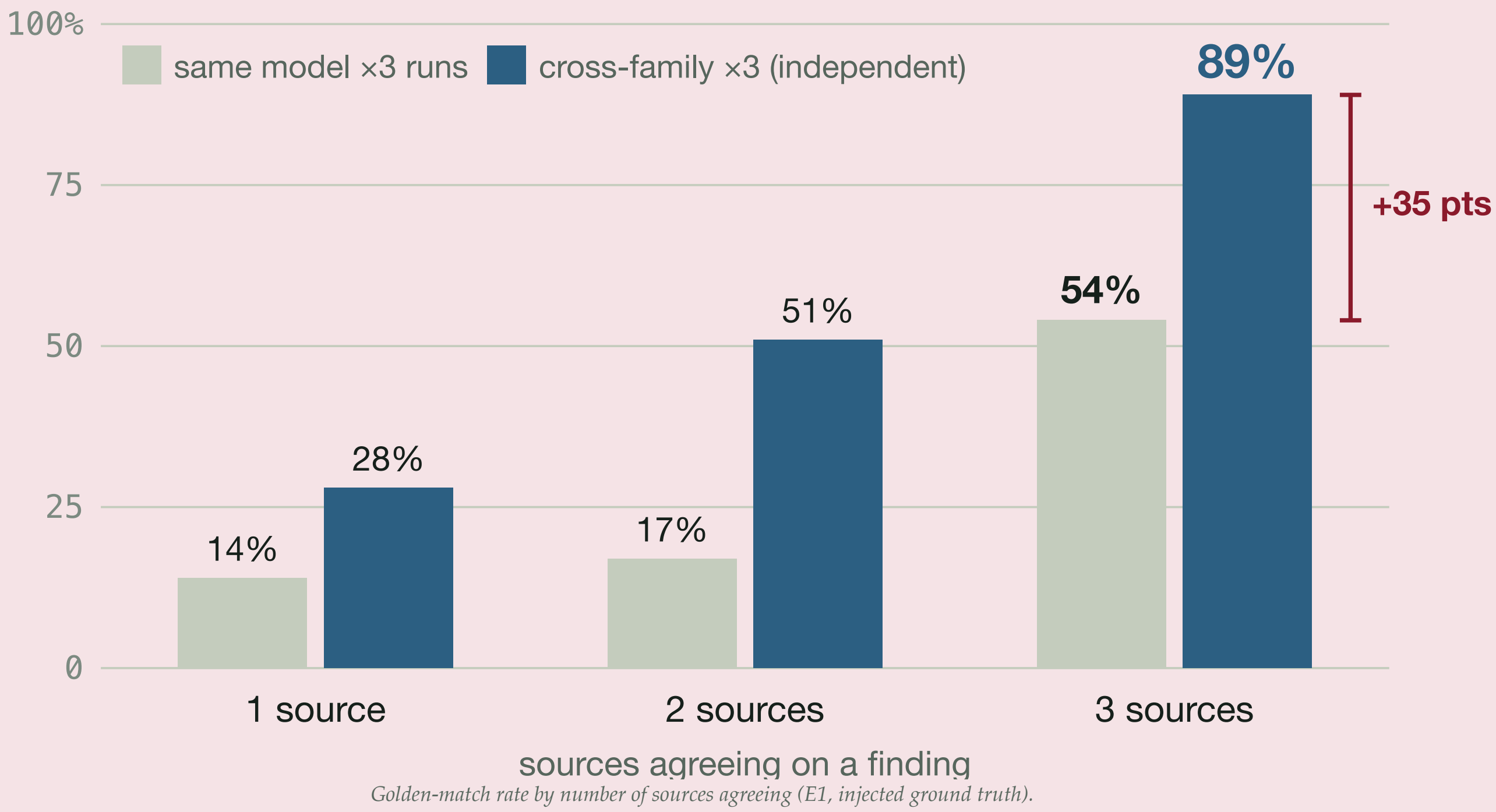
Error decorrelation — not more agents, specialists, or debate — is what turns agreement into a real signal.

89%

three **independent** model families that agree match injected ground truth

54%

one model run three times — agreement is mostly noise



01 PROBLEM & MOTIVATION

THE CLIENT

The **Indigenomics Institute** needs corporate Reconciliation Action Plan commitments made legible and trackable — and AI now writes much of the software that does it.

THE RESEARCH

Multi-agent AI code reviewers are proliferating — managers, specialists, voters — and priced accordingly. Does that **structure** improve review, or just cost more?

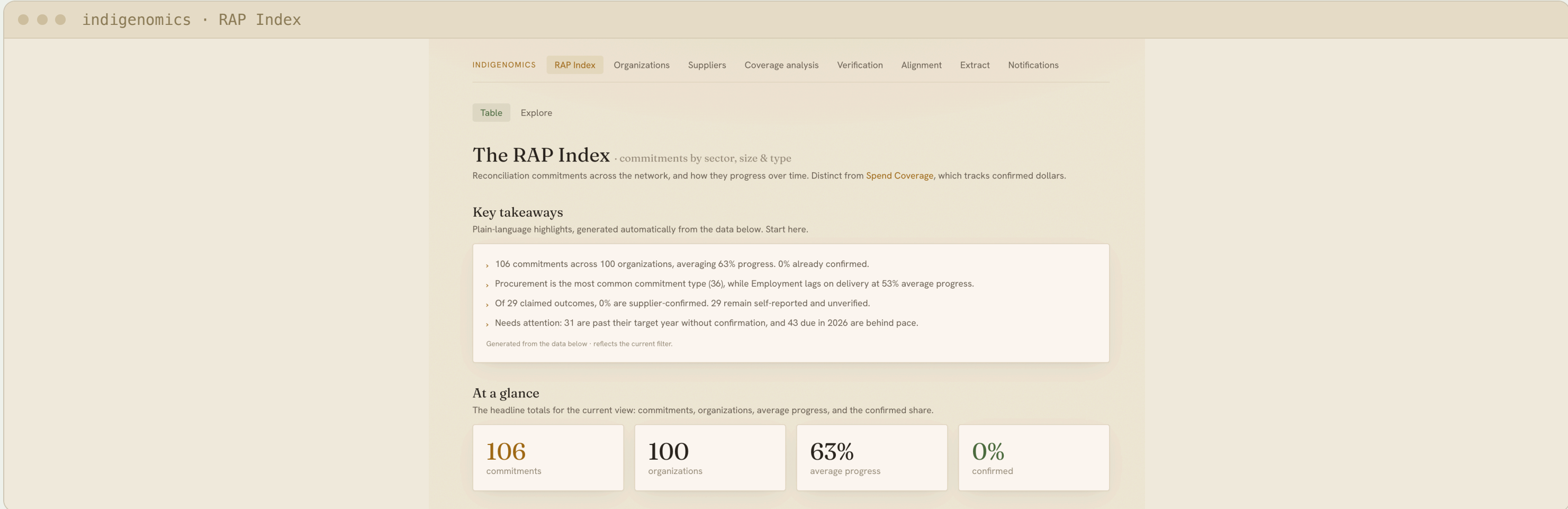
GROUNDING IN

Carol Anne Hilton's economic-reconciliation framework — an economic lens, not a case summarizer. Pre-registered on OSF; prompts frozen before any confirmatory data.

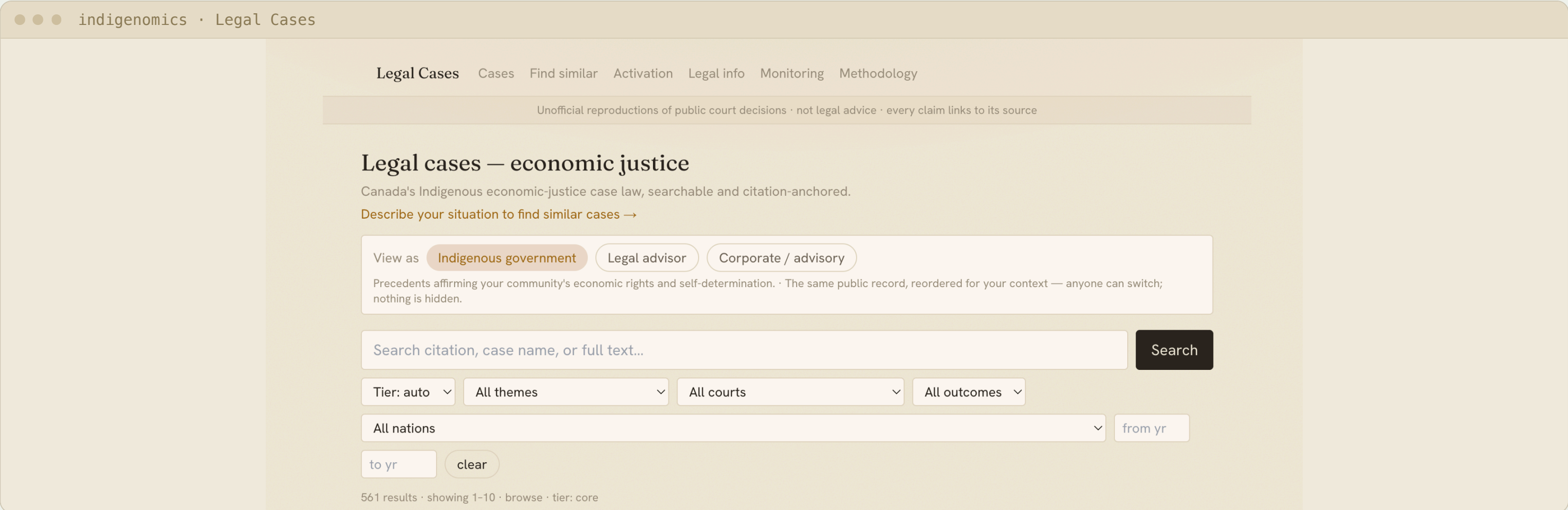
02 APPROACH

WHAT WE BUILT FOR THE CLIENT

The **RAP Data Portal** — one platform, two lenses: corporate reconciliation commitments and Indigenous legal precedents, as comparable searchable data on AWS.



Lens 1 — RAP Index. 106 commitments across 100 organizations, 63% average progress.



Lens 2 — Legal Cases. 561 Indigenous economic-justice decisions, citation-anchored.

THE EXPERIMENT — APPARATUS, NOT THE FINDING

Agentless

Generalists-3

Hierarchical

Consensus

Each rung adds exactly one capability. E3 is the portal's own 30 pull requests.

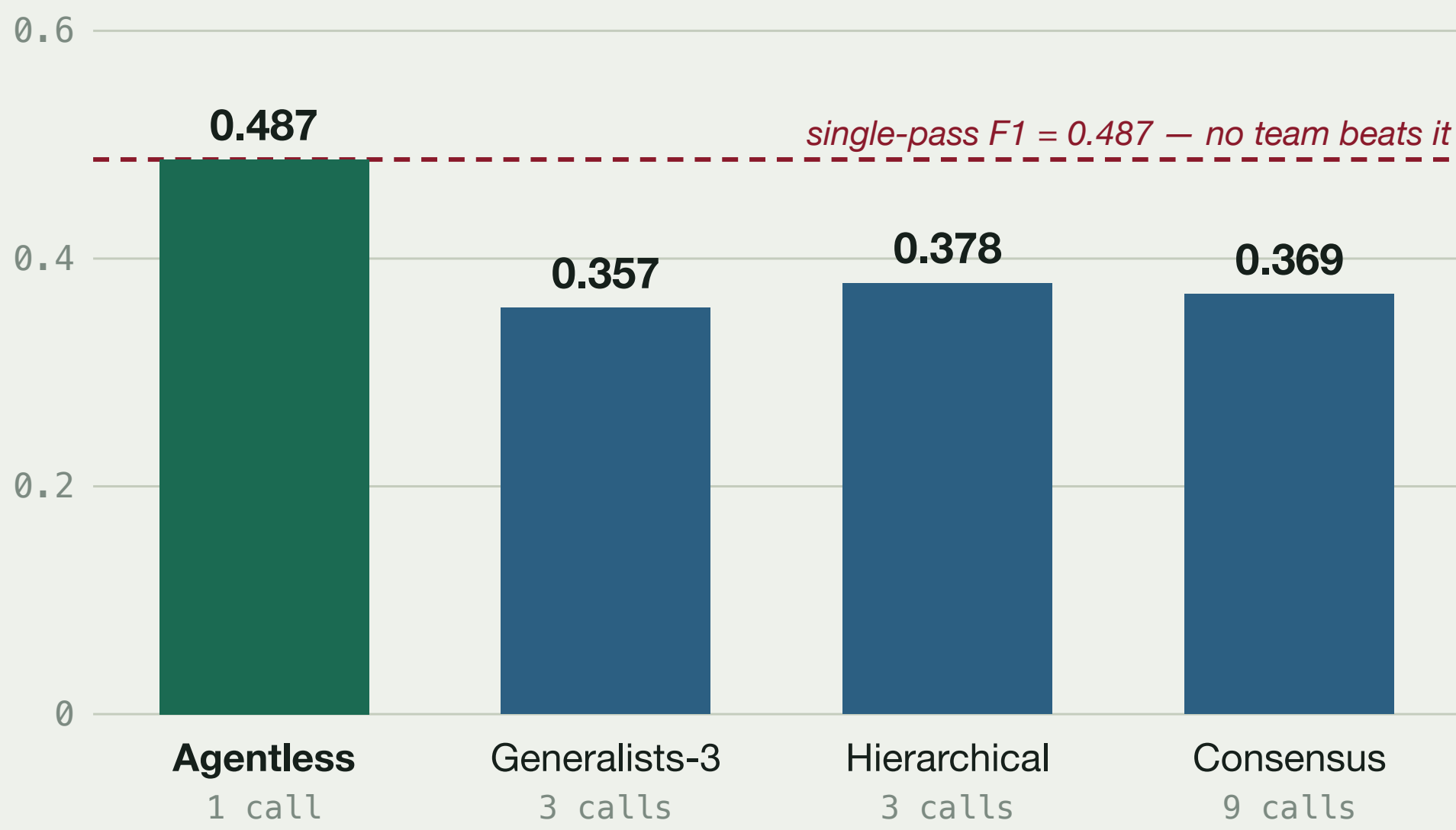
Held constant: same PR snapshot · same model (Claude Haiku 4.5) · same prompts · same evaluation pipeline · three runs per arm · **pre-registered on OSF**

03 KEY RESULTS

Structure is null. No multi-agent arm beats single-pass F1; specialists tie compute-matched sampling ($p=0.73$); peer debate adds nothing over a manager ($p=0.12$).

Decorrelation pays — on precision, not coverage (89% vs 54%).

Bounded. Recall plateaus at ≈ 0.83 ; on the portal's own unlabeled code, LLM correctness judges disagree ($\kappa=0.03$).



04 WHAT THE EXTRA AGENTS ACTUALLY BUY

E1 · Q000, 99 PRS

Agentless
1 call
0.34

Generalists-3
3 calls
0.45

Hierarchical
3 calls
0.50

Consensus
9 calls
1.76

5x more expensive per confirmed finding than single-pass review

Adding agents doesn't buy quality — it buys **recall at a price**. Every confirmed true finding costs **0.34** LLM calls under single-pass review and **1.76** under peer consensus, for a **lower-F1** result.

That is the practical case for decorrelation: **spend the budget on independence, not on more of the same reviewer.**

IMPACT — FIELD

Don't deploy multi-agent review expecting a quality lift. **Decorrelate your verifiers** — independent model families — and read their agreement as a precision signal.

IMPACT — CLIENT

A verifiable, trackable view of economic reconciliation — **commitments + legal precedents in one platform**, delivered to the Indigenomics Institute.

NEXT STEPS

Tool-grounded verifiers (execution + static analysis). **Open science:** all runs, ablations & artifacts released.